

Database Constraints

① Domain constraint

<u>ID</u>	name
01	Arib

id int;

INSERT <'ab', 'Rahman'>

⊗

Primary { NULL ⊗ — Entity integrity constraint
UNIQUE ✓ — Key

② Key constraint → Primary Key unique आ है

<u>ID</u>	name
1	a
2	b

<'2', 'Ahmed'> ⊗

<'3', 'b'> ✓

③ Entity integrity constraint

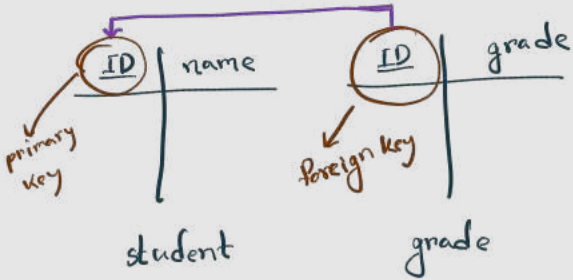
↳ Primary Key NULL

<u>ID</u>	name
NULL	

⊗

④ Referential Integrity constrain

F.K → Primary



(a)

<u>ID</u>	name
1	
2	

<u>ID</u>	Grade
1	
3	

* Foreign Key आ आ
Primary Key के 24 फल

⊗

(b)

<u>ID</u>	name
1	
2	
3	

<u>ID</u>	Grade
1	
2	

✓

(c)

<u>ID</u>	name
1	
2	
3	

<u>ID</u>	Grade
1	
2	

<u>ID</u>	name
1	
2	
3	

<u>ID</u>	Grade
1	
2	

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Yess, Houston, TX	M	40000	333445555	5
Alicia	J	Zelaya	999985777	1968-01-19	3321 Castle, Spring, TX	F	43000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	333445555	4
Ramesh	K	Namesh	666664444	1962-09-15	975 Fire Cak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-21	5631 Riisa, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987997897	1969-03-20	990 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888666555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888666555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666664444	3	40.0
453433453	1	20.0
453433453	2	20.0
333445555	3	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999987777	30	30.0
999987777	10	10.0
987987987	10	35.0
987997987	20	5.0
987654321	30	20.0
987654321	20	15.0
888666555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	3
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445559	Alice	F	1986-04-06	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

Suppose each of the following update operations is applied directly to the above database

① Insert <D1, Stanford>
in DEPT_Locations

Domain constraint
violate,

② Insert <6, Stanford>
in DEPT_Locations

Referential I.C
Reason: P.K @ 6 नहीं

③ Insert <null, Stanford>
in DEPT_Locations

Entity I.C

④ Delete Franklin row from employee table
Referential I.C

⑤ Update ProductX Pnumber to 20
Key constraint

How to **enforce** Database Constraints:

Constraints	Prevention / Solution / Enforce Constraint
1. Key Constraints	Set the attribute as Primary Key
2. Entity Integrity Constraints	Set the attribute as Primary Key
3. Referential Integrity Constrains (FK Constraint)	On Insert: Set proper Foreign Key with Primary key. On Update/Delete: Restrict -> operation does not run Cascade -> Delete rows where it matches Set Null -> Set Null where it matches
4. Domain Constraints	Set the attribute with proper DataType, set rules